

Sampling Completeness in Generative Search: Brand and Cited-Domain Accumulation under Repeated Queries

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Abstract

Whether repeated identical buying questions exhaust a language model’s brand recommendations depends on retrieval. Across 300 question–engine cells (50 questions, six engines, 15 runs each, open extraction over 1,470 adjudicated organizations), the five engines answering without web search were still adding never-seen brands at run 15 in 86–92% of cells, with median repertoires of 15–31 organizations; the one retrieval-enabled engine closed its list (median 8 organizations, 64% of cells still adding), matching four earlier deep cells where web-search runs saturated by run ten. Cited-domain accumulation keeps rising at every horizon tested. A single run shows 62–77% of the five-run brand set, and across engines the median question draws 38 organizations, of which a median of 15 appear in exactly one engine.

1 Accumulation in repeated answers

Auditors, brand trackers, and researchers sample LLM answers to commercial questions and report what they see. Every such report carries an implicit sampling claim about how much of what the model would say the collected runs have seen. Prior work established that identical prompts can yield different answers run to run [1, 5, 6], and that repetition inside a single prompt changes model behaviour [2]. What has not been measured is the accumulation structure of repeated answers: how fast the set of distinct brands, and the set of distinct cited domains, grows as runs accumulate.

We treat runs as sampling units and brands (or cited domains) as species, and apply the classical accumulation toolkit [3, 4]: the exact rarefaction curve $A(k)$, the expected number of distinct items after k of the n collected runs, averaged over run subsets; the final rarefaction increment $A(n) - A(n - 1) = Q_1/n$, where Q_1 counts items seen in exactly one run; and the Chao2 richness estimate, a lower bound, in its bias-corrected form when $Q_2 = 0$. Rarefaction interpolates within the collected sample; $Q_1 = 0$ means every observed item recurred, never that no unobserved item exists. Two studies follow: a first pass on existing corpora (June 2026 data) whose result the second, larger study revises.

2 Study 1: roster curves flatten; deep cells saturate under web search

We first reanalysed our public Category Ownership deposit [7]: 250 brand-free buying questions across five industries, put to GPT-5.2, Gemini 3 Flash, and Perplexity sonar-pro five times each (3,750 responses; of 750 cells, 737 hold responses, 705 hold at least four runs, 684 all five). Extraction used the deposit’s 50-brand alias roster. On that roster the median cell names 5 brands, one run shows 80% of the five-run set, and in 49% of complete cells at least one brand occurred in only one run.

Two deep probes (UK supermarkets; Nordic care providers; 24 runs each on GPT-5.4 and Gemini 3.5 Flash, both with web search) allowed an open check. Their stored rosters sat one brand under the observed plateaus, so we audited the extraction by open candidate mining; Table 1 reconciles the cells. With the extended rosters all four cells still reached $Q_1 = 0$, at 91–100% of the observed 24-run set by run ten, with repertoires of 7–11 organizations. Cited domains, recorded from the runs’ actual citations, kept accumulating: Q_1 of 18, 6, 41, and 12 at run 24, with 59–84% of the Chao2 estimate observed (Figure 2).

Table 1: Study 1 extraction reconciliation, deep cells.

Cell	Stored	Corrected	Entities added
UK / Gemini 3.5 Flash	9	11	M&S; Iceland (roster brand, missed)
UK / GPT-5.4	8	9	M&S
Nordic / Gemini 3.5 Flash	5	9	Ersta Diakoni, Blomsterfonden, Stora Sköndal, Bräcke Diakoni
Nordic / GPT-5.4	5	7	Ersta Diakoni, Aleris

Study 1 left open whether the brand plateau was a property of models or of the two designs that produced it: a fixed 50-brand roster at breadth, and web-search runs at depth.

3 Study 2: six engines, open extraction, and the retrieval split

We then collected 4,500 fresh responses in September 2026: the first 10 questions from each of the five industry banks (50 questions, verbatim), put 15 times each to six engines (GPT-5.6, Claude Sonnet 5, Gemini 3.7 Flash, Grok 4.5, Mistral Large, and Perplexity sonar, the one engine answering with web search). All 300 cells completed 15 of 15 runs with no errors. Extraction was open: capitalized candidate strings, pooled per question across engines, were adjudicated against one written rule (a candidate counts if it names an organization offering the queried category; product lines collapse to their organization) by a language-model classifier at temperature 0 and spot-checked by hand, yielding 1,470 organizations. A parallel roster-arm extraction reproduces Study 1’s flat breadth curves (median 4–5 brands per cell), so the earlier plateau belonged to the roster.

Three results (Figure 1, Table 2):

- **Without retrieval the brand list does not close by 15 runs.** On the five parametric engines, 86–92% of cells still added a never-seen organization at run 15; median repertoires run 15–31 organizations, and one run shows 62–72% of the five-run set. The late arrivals

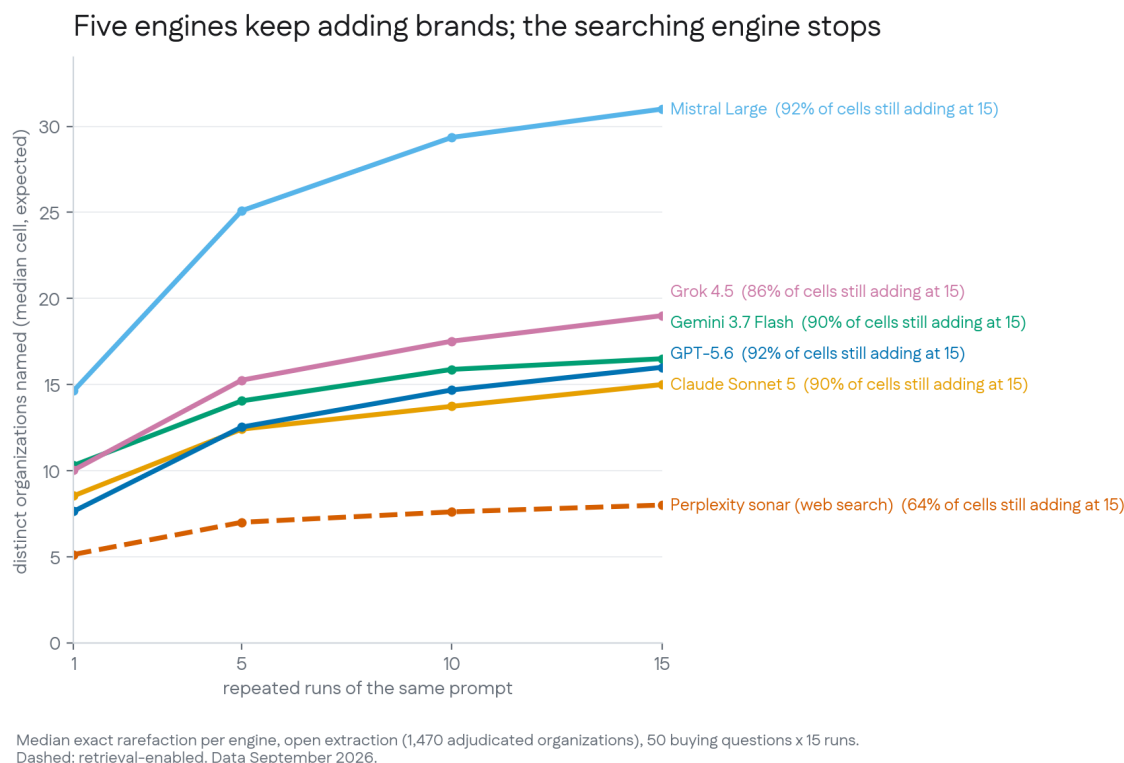


Figure 1: Median expected brand accumulation per engine under open extraction (50 buying questions, 15 runs per cell). The five engines answering without retrieval keep adding organizations through run 15; the retrieval-enabled engine levels off early. Percentages: share of that engine’s 50 cells with $Q_1 > 0$ at run 15.

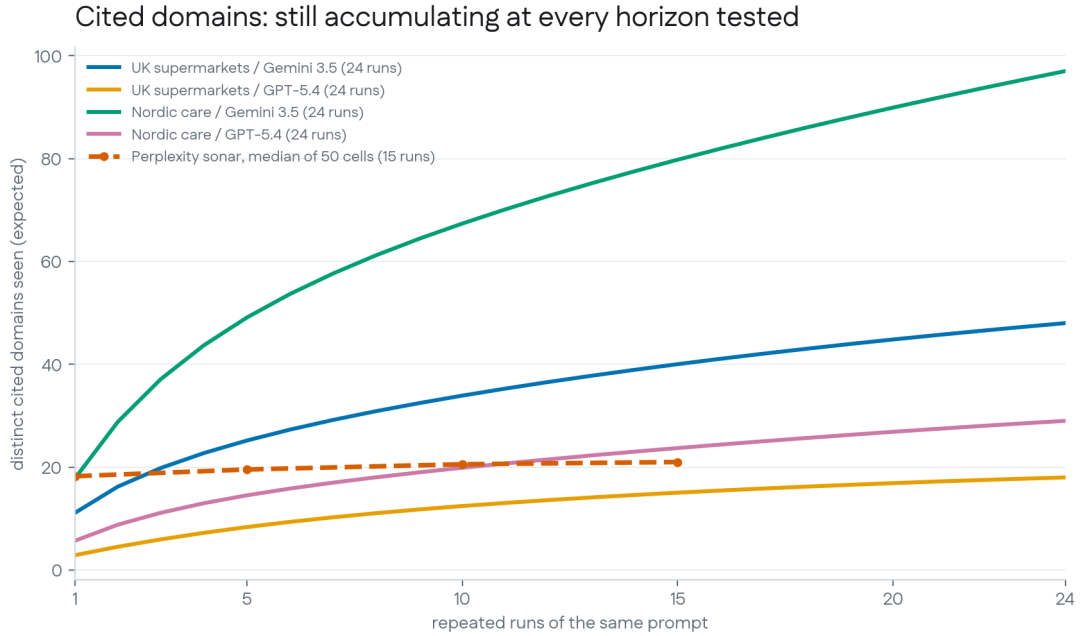
are genuine vendors (for the CRM question: Jobber, Copper, Nimble, Follow Up Boss), never extraction noise.

- **With retrieval it closes.** The web-searching engine’s median cell names 8 organizations, reaches 100% of its Chao2 estimate by run 15, and is the only engine where a majority of cells stop adding. This matches Study 1’s deep cells, both of which ran with web search, and identifies retrieval as the variable behind their saturation.
- **One engine is not the market.** The median question draws 38 organizations across the six engines; the best single engine shows 83% of that union (first quartile 74%), and a median of 15 organizations per question appear in exactly one engine.

The retrieval-enabled engine’s cited domains extend Study 1’s domain result to breadth: 44% of its 50 cells were still adding domains at run 15 (median 21 domains per cell, one run showing 88% of the 15-run set), against four deep cells all still rising at run 24 (Figure 2).

Table 2: Study 2 per engine, open extraction, medians over 50 cells.

Engine	Repertoire	Cells adding at 15	Seen at 15 (of Chao2)	$A(1)/A(5)$
Mistral Large	31	92%	94%	0.62
Grok 4.5	19	86%	91%	0.67
Gemini 3.7 Flash	16.5	90%	92%	0.72
GPT-5.6	16	92%	86%	0.65
Claude Sonnet 5	15	90%	88%	0.71
Perplexity sonar (search)	8	64%	100%	0.77



Exact rarefaction. Deep cells June 2026; sonar breadth September 2026.

Figure 2: Cited-domain accumulation: the four deep web-search cells of Study 1 (24 runs) and the Study 2 retrieval engine’s median cell (15 runs). Domain discovery continues at every horizon tested.

4 Related work

Non-determinism of repeated LLM queries is established [1], and our earlier work decomposes its variance and derives iteration requirements for stable means [5, 6]. This note measures set growth; mean stability is a different measurement with its own budgets. Leviathan et al. [2] show repetition inside one prompt improves accuracy; we measure repetition across calls. Citation instability in generative search is documented on single runs [8, 9]; the accumulation view quantifies how much of the citation record a sampling budget has observed. The estimators are the species-accumulation classics [3, 4].

5 Conclusion

Whether repeated queries exhaust an LLM’s brand recommendations is a property of the deployment, and retrieval is the switch. Grounded in web search, the answer set is short and a handful of repeats sees it whole. Ungrounded, five engines from five vendors were still producing never-seen organizations at the fifteenth identical run, with repertoires three to six times the grounded engine’s. Cited-domain accumulation, measured wherever citations exist, had not closed at any horizon tested. For anyone buying or selling repeated-query audits the budget question therefore splits three ways: grounded brand lists are cheap to finish; ungrounded brand lists and citation maps are open-ended at practical budgets; and none of these budgets settles recommendation frequencies, which need their own design [6]. Replication across categories, languages, decoding settings, and more retrieval configurations is the direct next step.

Declarations

Competing interests. The author is CEO of Rankfor.AI, which sells AI-visibility measurement, including repeated-query brand audits; this note bears on how such measurement should be priced and sampled.

Data and code. The Study 1 breadth corpus is our public deposit (Zenodo, 10.5281/zenodo.20788142, CC BY 4.0); the Study 2 collector, raw responses, adjudicated vocabulary, analysis code, per-cell tables, and the Study 1 extraction audit are at <https://github.com/Rankfor/rankfor-open/tree/main/research/recommendation-saturation>.

Provenance. Study 1 is a secondary analysis of the author’s own deposits; Study 2 was collected for this note; the accumulation analysis appears in no prior manuscript.

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